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RESEARCH

When to Re-Plan: Subgoal Persistence in Hierarchical Latent Reasoning

ACCEPTED · **Sole Author** · 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents @ ICML 2026 · *Seoul, South Korea*

Jan 2025 – May 2026 · *Paper* · *Code / SCOUT*

- Asked how long a hierarchical latent reasoner should preserve **medium-horizon intent** before revising it while multi-step computation unfolds in latent state.
- Extended HRM with a **feudal-style manager-worker interface**: a high-level module emits a directional latent subgoal that persists across worker steps, with hidden-state injection, commitment gating, and alignment supervision.
- Framed persistent subgoals as **reusable compositional structure for long-horizon planning**. Results showed that re-planning at every step can disrupt coherent structure, while moderate persistence guides computation without making the plan fixed; ablations separated learned direction from added capacity or auxiliary supervision.

A Score Is Not a Policy: Measuring Whether a Supervisory Module's Decisions Are Worth Making

Sole Author · Preprint · *Under Review*

May 2026 – Present · *Paper*

- Asked whether a learned supervisor can use a reasoner's internal state to decide when an active commitment should persist or be revised, moving from fixed schedules to **state-conditioned meta-control**.
- Developed an evaluation framework separating **score dependence, behavioural adaptation, and decision value**, using strong fixed schedules, matched PERSIST / REPLAN counterfactuals, controlled timing interventions, and oracle comparisons.
- Found that an apparently adaptive score need not produce a useful policy: simple timing structure explained much of the controller's behaviour, while strong fixed strategies captured much of the available value. The result motivates stricter evaluation of supervisory modules in compound agentic systems.

SAINT

Independent Research Project 2026 – Present

- Selected for **Accel AI Innovate: Paris**, Accel's Future Founders hackathon for AI builders, hosted with Iterate (October 2026).
- Participating in the **SpaceXAI Macau Community Hackathon** at Macau Tower, Macao, in Track 2: **Physical AI** × **Grok Bot** (November 2026).
- **Current direction – reasoning allocation:** studying how an agent can estimate whether additional computation, another attempt, or a change in strategy is worth its cost, with planned experiments in coding agents working across real software repositories.
- Extends earlier work on **subgoal persistence** and **supervisory decision value** from deciding when to revise an internal commitment to allocating reasoning effort more broadly.

RESEARCH SERVICE

Invited Reviewer · NeurIPS 2026 Workshop on Responsible Use of Meta-Agents

Nominated by the Area Chair to review work on agents that design, optimize, supervise, train, or evaluate other agents, including agent harnesses and meta-level evaluation.

Reviewer · 2nd Workshop on Compositional Learning: Safety, Interpretability, and Agents @ ICML 2026

Reviewed research on compositional learning, reasoning, interpretability, and agentic systems.

PROFESSIONAL EXPERIENCE

Propero Consulting
Pune, Maharashtra, India

Software Developer · Machine Learning & Applied AI Aug 2022 – Dec 2024

- **Led the design and engineering of ShopiBot**, a domain-specific AI system supporting Shopify developers with technical questions, client requirements, and implementation guidance; translated an early product idea into retrieval, orchestration, and evaluation workflows.
- Studied and implemented retrieval approaches including dense semantic search, **RAPTOR-inspired hierarchical retrieval**, and corrective and adaptive RAG; combined query analysis with routing across retrieval pipelines to improve access to Shopify technical context.
- Built agentic workflows with knowledge graphs, LangChain planners, multi-tool reasoning, self-reflection, and LLM tool calling; explored **Sema4AI / Robocorp** for developer automation and created technical demos for prospective startup partnerships.
- Improved production reliability through automated evaluation, streaming inference, vector-database sharding, and caching, reducing end-to-end latency from **53s to 20s**. Earlier work included CLIP-guided generation and automated checkout-flow testing using DOM analysis and feedback-driven methods.

Unify · YC-backed
London, UK / Remote

LLM Engineer Contributor Jul 2024 – Sep 2024

- Selected for Unify's contributor programme, collaborating with the core team on a community-built **unified LLM query router** across model providers.

EDUCATION

Northeastern University
Boston, MA

M.S. Artificial Intelligence · Machine Learning concentration Admitted Fall 2026

Khoury College of Computer Sciences · Spring 2027 start option

International Impact Award · Inaugural Cohort Scholarship · 30% tuition scholarship

College of Technology, GBPUA&T
Pantnagar, India

B.Tech, Electrical Engineering · 73.2% · First Division 2017 – 2021

TECHNICAL SKILLS

Reasoning & ML: Hierarchical and recurrent latent reasoning, *adaptive computation*, supervisory / meta-level control, test-time computation, hierarchical and feudal RL, PyTorch, Hugging Face Transformers

LLM & Agentic Systems: Dense retrieval, RAPTOR, corrective/adaptive RAG, LangChain, LlamaIndex, tool calling, LangSmith, CLIP, diffusion models

Engineering & Infrastructure: Python, C, C++, CUDA, Docker, AWS (S3, Lambda, EKS), Pinecone, Chroma, mixed precision, gradient checkpointing, single-node multi-GPU training